DSE Sample Paper CHEM P1A

SECTION A

PART I

1. **B**

Method 1

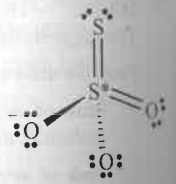
Since 1 molecule of decane gives 2 molecules of propene and 1 molecule of *X* upon cracking and the molecular formula of propene is C_3H_6, there are in total $10 - 2 \times 3 = 4$ carbon atoms and $22 - 2 \times 6 = 10$ hydrogen atoms in *X*. *X* is C_4H_{10}.

Method 2

There are in total $10 - 2 \times 3 = 4$ carbon atoms in *X*. As the molecular formula of alkanes in the homologous series can be represented by C_nH_{2n+2}, *X* should be C_4H_{10}.

2. **A**

According to the three-dimensional structure of $S_2O_3^{2-}$, the central S uses two of its valence electrons to form the S=O bond, two to the form S=S bond, and one for each of the S–O bonds.



+2 −2 +1 −1
S*=O S*–O⁻
0 0
S*=S

Now, when you are to assign the oxidation state for the central S, as O is always more electronegative, the bonding electrons will shift to O. As a result, the central S will lose four valence electrons to O.

Finally, for the S=S unit, as they have the same electronegativitiy, then the four bonding electrons (two from each S) will be equally shared by the two S atoms. It means that the central S bears two valence electrons while the corner S atom has six (it already has two lone pairs).

Therefore, the oxidation state of the central S atom is $6 - 2 = +4$ while that of the corner S atom is $6 - 6 = 0$.

For SO_4^{2-}: Let x be the oxidation number of S.

$x + (-2) \times 4 = -2 \quad \therefore \quad x = +6$

For HSO_3^-: Let x be the oxidation number of S.

$x + (-2) \times 3 + (+1) = -1 \quad \therefore \quad x = +4$

For $S_2O_7^{2-}$: Let x be the oxidation number of S.

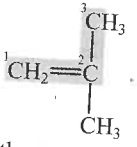
$2x + (-2) \times 7 = -2 \quad \therefore \quad x = +6$

$\therefore$ Sulphur exhibits the lowest oxidation number in $Na_2S_2O_3$.

3. **B**

The insoluble sand suspension in the mixture can be filtered as the residue, since the sand particles are too large to pass through the pores of the filter paper. Then, distillation is adopted to separate water and salt in the filtrate (i.e. salt solution). Note that evaporation is not used to separate water and salt as this would result in a loss of the solvent water.

4. **C**

The monomer of X should be 

The longest carbon chain of the monomer consists of three carbon atoms and hence X is propene. With methyl group substituted at the 2nd C, the systematic name of X should be methylpropene.

Note: It is not necessary to indicate the position of the methyl group as '2-methylpropene'. If the methyl group were substituted at the 1st C, X would be but-2-ene. If the methyl group were substituted at the 3rd C, X would be but-1-ene.

5. **D**

At the initial stage of rusting, Fe would be oxidised by atmospheric oxygen to give $Fe^{2+}(aq)$ in the presence of both oxygen and water.

Oxidation: $Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$

Reduction: $2H_2O(l) + O_2(g) + 4e^- \rightarrow 4OH^-(aq)$

The rust indicator can serve as a sensitive test for $Fe^{2+}(aq)$, turning blue itself.

6. **C**

As magnesium and zinc are more reactive metals than iron, they would sacrificially protect iron from rusting. Hence, iron nails wrapped by magnesium or zinc strips would not rust. The opposite situation occurs for dish 1 and dish 3, in which the iron nails sacrifice themselves to protect the silver strip and the copper strip.

7. **A**

The atomic number of an atom refers to the number of protons in the atom while the mass number refers to the sum of the number of protons and neutrons in the atom.

No. of protons in the atom = 18

No. of neutrons in the atom = 40 − 18 = 22

Since the atom is electrically neutral, no. of electrons = no. of protons = 18.

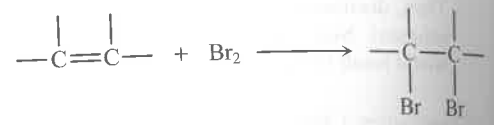
8. **D**

The acid, as shown by the labels, is corrosive and oxidising.

Note that a very concentrated acid is not necessarily corrosive. Concentrated sulphuric acid is corrosive, owing to its dehydrating property, but most concentrated weak acids are not corrosive, for instance, concentrated ethanoic acid.

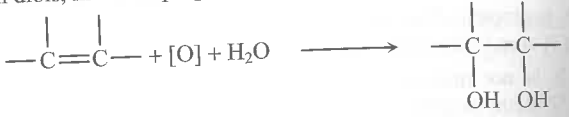
9. **B**

A ✓ Alkenes undergo addition reaction with bromine to decolourise bromine solution.

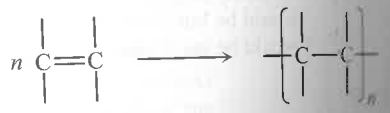


B × Alkenes cannot serve as a bleaching agent so it would not be able to decolourise red litmus solution.

C ✓ Alkenes can be oxidised by acidified potassium permanganate solution to form diols, in which purple MnO_4^- is reduced to form pale pink Mn^{2+}.



D ✓ Alkenes can be polymerised to form poly(alkenes) owing to the presence of the unsaturated C=C bonds.



10. **D**

A × It is a displacement (spontaneous redox) reaction, which is exothermic.

B × It is a neutralisation reaction, which is exothermic.

C × It is a combustion reaction, which is exothermic.

D ✓ It is a cracking reaction, which is endothermic.
Cracking involves breaking of long-chain hydrocarbons into short ones like small alkanes and alkenes.

11. **C**

Isotope	^{206}X	^{207}X	^{208}X
Relative abundance	22%	23%	55%

$\therefore$ The relative atomic mass of $X = 206 \times 22\% + 207 \times 23\% + 208 \times 55\% = 207.3$

12. **B**

Since copper is at a higher position in the electrochemical series than silver, it is a stronger reducing agent. Then oxidation takes place at the copper rod while reduction takes place at the silver rod. Hence, the copper plate becomes the negative electrode (anode) while silver becomes the positive electrode.

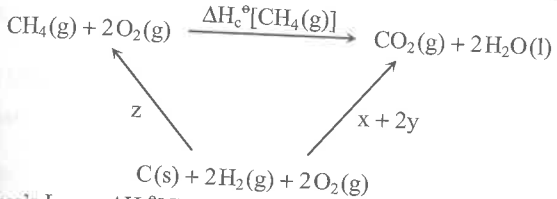
The following chemical reactions would take place:

At copper plate: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$

At silver plate: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

The mass of the anode would decrease as the copper rod dissolves in the solution. However, the concentration of $Cu^{2+}(aq)$ in the solution remains unchanged since 1 mole of Cu(s) would be oxidized at the anode for 1 mole of $Cu^{2+}(aq)$ reduced at the cathode. The colour intensity of the solution would remain unchanged.

13. **C**



By Hess's Law, $\Delta H_c^\circ[CH_4(g)] = x + 2y - z$

Note: x is the standard enthalpy change of formation of $CO_2(g)$.
y is the standard enthalpy change of formation of $H_2O(l)$.
z is the standard enthalpy change of formation of $CH_4(g)$.

14. **B**

No. of moles of $Ca(OH)_2$ in the solution $= \dfrac{3.7}{40.1 + 16.0 \times 2 + 1.0 \times 2} = 0.0499325$

Molarity of the solution $= \dfrac{0.0499325}{500 \times 10^{-3}} = 0.0998650\ M \approx 0.10\ M$

15. **A**

Number of molecules = Number of moles × Avogadro constant

$$= \frac{\text{Mass}}{\text{Molar Mass}} \times \text{Avogadro constant}$$

Since the samples have the same mass (10 g), the one containing the smallest number of molecules is the one having the largest molar mass.

For NO_2: Molar mass $= 14.0 + 16.0 \times 2 = 46.0$
For CO_2: Molar mass $= 11.0 + 16.0 \times 2 = 44.0$
For H_2S: Molar mass $= 1.0 \times 2 + 32.1 = 34.1$
For C_2H_4: Molar mass $= 12.0 \times 2 + 1.0 \times 4 = 28.0$
$\therefore$ NO_2 contains the smallest number of molecules.

16. **A**

A 25.0 cm^3 pipette is used to measure and extract 25.0 cm^3 of the toilet cleaner.

17. **A**

Since concentrated sulphuric acid present in the solution is titrated with sodium hydroxide solution, pH of the reaction mixture is increasing (i.e. becomes more alkaline) during the course of the experiment. Phenolphthalein turns from colourless to pink at the end-point.

18. **C**

No. of moles of NaOH used in the titration $= 0.950 \times (27.1 \times 10^{-3}) = 0.025745$

$$H_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$$

$\therefore$ Mole ratio of H_2SO_4 : NaOH is 1 : 2.

No. of moles of H_2SO_4 in the 25.0 cm^3 diluted sample $= 0.025745 \times \frac{1}{2} = 0.0128725$

No. of moles of H_2SO_4 in the undiluted sample $= 0.0128725 \times \frac{250.0}{25.0} = 0.128725$

Concentration of sulphuric acid in the undiluted sample $= \frac{0.128725}{25.0 \times 10^{-3}} = 5.149\ M \approx 5.15\ M$

19. **D**

No. of moles of KCl(s) dissolved $= \frac{10.0}{39.1 + 35.5} = 0.134048$

Energy absorbed by the mixture $= m_{total} c \Delta T$
$= (m_{KCl} + \rho_w V_w) c \Delta T$
$= (10.0 + 1.0 \times 100)(4.2)(5.5)$
$= 25410\ J$
$= 2.541\ kJ$

Molar enthalpy change of the dissolving process $= \frac{2.541}{0.134048} = 18.96\ kJ\,mol^{-1}$

Note: The mass of salt added is dominant as the volume of water added is not too much.

20. **C**

(1) ✓ Dilute sulphuric acid would not react with carbon but it can neutralize CuO. The reaction mixture would warm up due to the highly exothermic neutralization reaction only if the black powder contains copper(II) oxide.

(2) × Since both C and CuO would not react with NaOH, the black powder cannot be identified by the addition of NaOH.

(3) ✓ Since C can reduce CuO to Cu, the black powder may turn reddish brown upon strong heating. If the black powder comprises carbon only, the solid would remain black upon heating.

21. **D**

(1) ✓ Although Cu^{2+} is at a lower position in the electrochemical series, H^+ is discharged at the beginning because, at first, the concentration of Cu^{2+} ions around electrode *X* is much less than that of H^+ ions.

(2) ✓ Since positively charged Cu^{2+}, which is blue in colour, would be attracted to the negative electrode *X* while negatively charged $Cr_2O_7^{2-}$, which is orange in colour, would be attracted to the positively charged electrode *Y*. Hence, orange colour would appear in the solution around electrode *Y*.

(3) ✓ It can be observed that blue colour appears in the solution around electrode *X* while orange colour appears in the solution around electrode *Y*. This shows that ions migrate towards oppositely charged electrodes.

22. **D**

(1) × I_2 exhibits a simple molecular structure instead of a giant covalent structure.

(2) ✓ I_2 molecules are held together by weak van der Waals' force owing to the instantaneous dipole-induced dipole interaction.

(3) ✓ Iodine exists as diatomic molecules in which each I atom shares 1 electron to form the covalent bond.

23. **A**

1st statement: ✓

SO_3^{2-}, which is a reducing agent, can reduce bromine but sulphate cannot, since S in SO_4^{2-} has already achieved its highest oxidation state (+6), i.e. it cannot be further oxidized by bromine.

$$SO_3^{2-}(aq) + Br_2(l) + H_2O(l) \rightarrow SO_4^{2-}(aq) + 2Br^-(aq) + 2H^+(aq)$$

2nd statement: ✓

(See the explanation of the **1st statement**.)

24. **C**

1st statement: ×

CO_2 (a gas at room temperature) exhibits a simple molecular structure while SiO_2 (a solid at room temperature) exhibits a giant covalent structure. Hence, they do not share similar physical properties.

2nd statement: ✓

Both C and Si belong to Group IV, so each of their atoms has 4 outermost shell electrons. The atoms of elements in the same group have the same number of outermost shell electrons.

PART II

25. **A**

A × The molarity of HCl in the system remains unchanged so the rate of reaction would also remain unchanged.

B ✓ As the concentration of reactants is doubled, the chance of collisions between molecules increases. This increases the number of effective collisions and thus the rate of reaction increases.

C ✓ Reaction rate depends on the concentration of reactants in the system instead of the volume of solution.

D ✓ Using calcium carbonate powder would provide a larger surface for collisions to take place. The chance of collisions is higher and the number of effective collisions increases so is the rate of reaction.

26. **C**

A × This is an addition reaction.

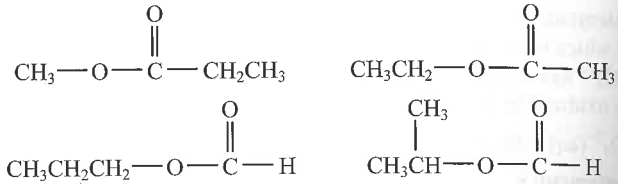
B × This is an oxidation reaction.

C ✓ A substitution reaction is a chemical change in which an atom (or a group of atoms) of an organic molecule is replaced by another atom (or a group of atoms).
In this reaction, a –OH group is substituted by a –Br group.

D × This is a reduction reaction.

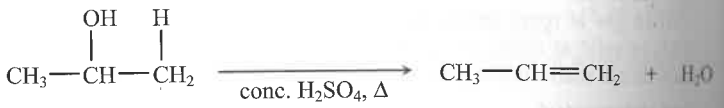
27. **B**

There are in total 4 possible structures of $C_4H_8O_2$ with an ester functional group.



28. **C**

The conversion of propan-2-ol to propene is a dehydration reaction, since a H_2O molecule is eliminated.



29. **C**

If the temperature of the system is lowered, both the rates of forward and backward reactions would decrease since the kinetic energy of the reactants would decrease, and hence there is a smaller fraction of molecules having enough kinetic energy for effective collisions. Thus the rate of reaction decreases.

30. **A**

A ✓ Combustion of hydrogen produces water only, which is friendly to the environment but incomplete combustion of butane may produce air pollutants such as air particulates and carbon monoxide, as well as carbon dioxide that leads to greenhouse effect and results in global warming eventually.

B × Since the carbon content and C : H ratio of butane are higher than that of methane, butane is harder to burn completely than methane. Hence, butane would burn with a more sooty flame.

C × Methane and butane belong to the same homologous series, alkane, which can be represented by the structural formula C_nH_{2n+2} while hydrogen is not an alkane.

D × $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

$C_4H_{10}(g) + \frac{13}{2}O_2(g) \rightarrow 4CO_2(g) + 5H_2O(l)$

On complete combustion, 1 mole of methane would release 1 mole of CO_2 while 1 mole of butane would release 4 moles of CO_2. Hence, 1 mole of butane would release more CO_2 than 1 mole of methane on complete combustion.

Note that the number of mole of CO_2 produced equals the number of mole of C in the fuel.

C

If reagent **A** is an oxidising agent, ethanol is oxidised to ethanoic acid which would react with the carbonate. If a dehydrating agent is used, ethene is formed which does not react with the carbonate.

Thus the first reaction is given by

$CH_3CH_2OH + 2[O] \rightarrow CH_3COOH + H_2O$

Thus **P** is ethanoic acid. Then the second reaction is given by

$2CH_3COOH(aq) + Na_2CO_3(aq) \rightarrow 2CH_3COONa(aq) + CO_2(g) + H_2O(l)$

Thus product **Q** is sodium ethanoate.

B

Concentration of $S_2O_3^{2-}$ in $W = \frac{1.0 \times 80}{90} = 0.889$ M

Concentration of $S_2O_3^{2-}$ in $X = \frac{1.5 \times 60}{90} = 1.0$ M

Concentration of $S_2O_3^{2-}$ in $Y = \frac{2.5 \times 30}{90} = 0.833$ M

Concentration of $S_2O_3^{2-}$ in $Z = \frac{3.0 \times 20}{90} = 0.667$ M

Since the concentration of $S_2O_3^{2-}$ in conical flask X is the highest, the reaction in conical flask X should proceed at the fastest rate.

C

$S_2O_3^{2-}(aq) + 2H^+(aq) \rightarrow SO_2(g) + S(s) + H_2O(l)$

(1) × Although gaseous SO_2 is given out in the reaction, it cannot be collected with a syringe as it dissolves readily in aqueous solutions.

(2) ✓ Since S(s) is formed, the rate of reaction can be determined by the rate of disappearance of the cross marked on the paper placed under the conical flask containing the reaction mixture. Hence, a stop watch is required to take the time required for the disappearance of the cross.

(3) ✓ A measuring cylinder is used to measure the different volumes of solution required.

34. **D**

(1) × Soap is derived from salts of carboxylic acids and alkanals but not alkenes.

(2) ✓ Owing to the presence of C=C double bonds and the short carbon chain length, the first few members would undergo addition polymerisation in the presence of free radical initiator to give polymers.

(3) ✓ Hydrohalogenation would take place such that the hydrogen halides would have electrophilic addition on alkenes to yield haloalkanes.

35. **B**

1st statement: ✓

Catalysts are used in processes like Haber Process to produce NH_3 and Contact Process to produce H_2SO_4.

2nd statement: ✓

The **2nd statement** is not a valid explanation of the **1st statement**. The real reason is that a catalyst speeds up the rate of reaction and lowers the temperature required. Thus the production processes can be conducted in a more time-efficient and cost-efficient way.

Note: The use of catalysts in industrial processes is discussed in detail in the elective part (Topic XIII Industrial Chemistry).

36. **C**

1st statement: ×

Neutralization would not take place since ethanol is slightly acidic and ethanoic acid is acidic too. No product would be given.

2nd statement: ✓

Esterification is a condensation process, in which H_2O is produced.

$$CH_3COOH + CH_3CH_2OH \xrightarrow[\text{conc. } H_2SO_4,\ \Delta]{} CH_3COOCH_2CH_3 + H_2O$$

END

DSE Sample Paper CHEM P1B

SECTION B

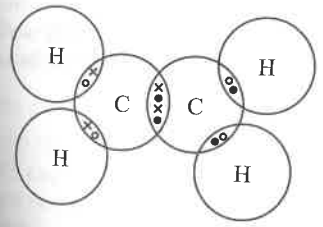
PART I

1. (a) False. Since the non-directional electrostatic attraction between Na^+ and Cl^- in NaCl is stronger than the weak van der Waals' force holding the methane molecules, the melting point of sodium chloride is much higher than that of methane. In fact, the covalent bonds between the atoms in methane molecules (C–C, C–H bonds) are stronger than the ionic bonds within sodium chloride lattices.

(b) False. The concentrated sulphuric acid should be added to the water slowly upon dilution. As the dilution of concentrated sulphuric acid is highly exothermic, the concentrated sulphuric acid may splash out due to heat evolved when water is added slowly to the concentrated sulphuric acid. The slow addition of concentrated sulphuric acid to water can allow the even spreading of heat as water is a good coolant.

(c) False. A is a stronger acid than B due to the greater extent of ionization of A in water than B to give $H^+(aq)$. The pH of the solution depends on the concentration of $H^+(aq)$ in the solution in addition to degree of ionization of acid. Hence, a concentrated solution of the weaker acid B may have a lower pH than a diluted solution of the stronger acid A. pH of two acids can only be compared when their strengths and concentrations are known.

2. (a)



(b) addition polymerization

(c) (Any **ONE** of the following)
- Polyethene is flexible.
- Polyethene is light.
- Polyethene is chemically unreactive.
- Polyethene is water-proof.
- Polyethene has a high tensile strength.
- Polyethene has a low density.

(d) (i) (Any **ONE** of the following)
- Landfilling
- Incineration
- Recycling

(ii) (Landfilling)

Advantage: (Any **ONE** of the following)
- It is simple.
- The cost is low.
- Landfilling does not produce toxic gases which causes air pollution.
- Methane produced can be used to generate power.

Disdvantage: (Any **ONE** of the following)
- Most plastics are non-biodegradable and chemically inert. They cannot be decomposed even for hundreds of years.
- It is impractical when there is a shortage of landfill site.
- Underground water may also be polluted.

(Incineration)

Advantage: (Any **ONE** of the following)
- Incineration can greatly reduce the volume of wastes.
- Requires less land for landfill.

Disdvantage: (Any **ONE** of the following)
- Burning of polyethene produces carbon dioxide causing greenhouse effect.
- Incomplete combustion of polyethene produces carbon particles, carbon monoxide, particulates and unburned hydrocarbons as air pollutants.

(Recycling)

Advantage: (Any **ONE** of the following)
- Saves resources like petroleum.
- Reduces the volume of solid waste.
- Less air pollutants are produced.

Disdvantage: (Any **ONE** of the following)
- Sorting out different plastics is time-consuming.
- It involves a high transport cost.

3. (a) Hydrogen gas was formed when *X* is added to cold water.
The gas gives a 'pop' sound when a burning splint is applied.

(b) redox reaction

Explanation:
Y undergoes reduction as its oxidation number changes from a positive value to zero when the oxide of Y is reduced to its metal. At the same time, carbon is oxidized to carbon dioxide and its oxidation number changes from 0 to +4. Hence, it is a redox reaction.

(c) Increasing reactivity: $Z < Y < X$

Only metal *X* among the three can react with cold water to give hydrogen gas, therefore only *X* is at a higher position than hydrogen in the electrochemical series. Also, only metal *X* cannot be reduced by carbon, so it is the most reactive one among the three.

Metal *Y* undergoes displacement reaction with copper(II) sulphate, so *Y* is a stronger reducing agent than copper. It is at a position higher than that of copper in the electrochemical series.

Since metal *Z* cannot displace the copper(II) ions, so metal *Z* is at a position lower than copper in the electrochemical series.

(d) There are water molecules $H_2O(l)$ in the copper(II) sulphate solution. *X* reacts with water to give colourless hydrogen gas.

4. (a) (i) The colour of the liquid changed from green to blue.

Both hydrogen ions and sodium ions moved towards rod *A* during electrolysis. However, $H^+(aq)$ ions would be preferentially discharged as $H^+(aq)$ ions occupy a lower position than $Na^+(aq)$ ions in the electrochemical series.

At rod *A*, $2H^+(aq) + 2e^- \rightarrow H_2(g)$

Since the concentration of $H^+(aq)$ near rod *A* decreased and became lower than that of $OH^-(aq)$, the solution became alkaline and hence the colour of universal indicator turned blue.

(ii) The gas was oxygen.

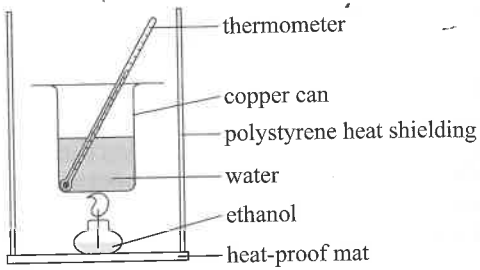
Both hydroxide and sulphate ions moved towards rod *B*. However, $OH^-(aq)$ ions would be preferentially discharged to oxygen as $OH^-(aq)$ ions occupy a higher position than $SO_4^{2-}(aq)$ ions in the electrochemical series.

At rod *B*, $4OH^-(aq) \rightarrow O_2(g) + 2H_2O(l) + 4e^-$

(b) pencils

Explanation: A pencil contains a graphite stick.

5. (a)



(b) Heat released from the combustion of ethanol $= mc\Delta T$

$= 200.0 \times 4.2 \times 6.0$

$= 5040$ J

$= 5.04$ kJ

No. of moles of ethanol (CH_3CH_2OH) burnt $= \dfrac{0.185}{12.0\times2+1.0\times6+16.0}$

$= 4.02 \times 10^{-3}$

Enthalpy change of combustion of ethanol $= \dfrac{-5.04}{4.02\times10^{-3}}$

$= -1254\ \text{kJ mol}^{-1}$

(c) (Any **ONE** of the following)
- There is no heat loss to the surroundings.
- There is complete combustion of ethanol.
- The heat transfer is complete.

6. (a) As O is more electronegative than H in the O–H bond of H_2O, the bonding electron cloud is drawn closer to O. O would acquire a partial negative charge (δ–) while H would acquire a partial positive charge (δ+). Hence, O–H bond in H_2O is polar.

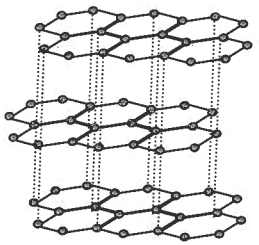
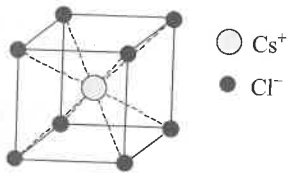
Besides, H_2O is V-shaped owing to the presence of lone pair electrons in O. The dipole moment of O–H bonds cannot cancel out each other and there is net dipole moment in H_2O molecule.

The jet of water was deflected by the negatively charged rod owing to the electrostatic attraction between the negative charges on the rod and the δ+ in H of H_2O.

(b) The jet of water would still be deflected towards the rod if the negatively charged rod is replaced by a positively charged rod since the water molecules would orient themselves with O facing the positive charges. The electrostatic attraction between δ– of O and the positive charges causes the deflection.

(c) The liquid jet would not be deflected since hexane is non-polar. Although weak dipole is induced under electric field, the induced dipole is not strong enough to cause deflection.

7.

Solid substance	Three-dimensional diagram for the structure of the solid substance	Explanation of whether the solid substance is an electrical conductor
diamond		Diamond would not conduct electricity since each carbon atom forms four covalent bonds with other carbon atoms, and all the valance electrons are localised in covalent bonds. There are no delocalised electrons or mobile ions to carry electric charge.
graphite		Graphite would conduct electricity since carbon atoms in graphite only form three bonds with others. The unpaired electrons are delocalised within the layers of carbon atoms. The delocalised electrons are free to move from one 6-membered C ring to the next within a layer. The delocalised electrons can help to conduct electricity.
caesium chloride		Caesium chloride in solid state is not a conductor of electricity because the Cs^+, Cl^- are not mobile owing to the package of the ions into the compact crystal lattice. There is strong ionic attraction between the ions, restricting their movements and the KE of ions are low at solid state.

8. (a) There is effervescence of colourless gas bubbles giving a colourless resultant solution.

$$Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$$

(b) A dirty green gelatinous precipitate would be formed owing to the formation of insoluble $Fe(OH)_2$. The solution resulted will be colourless.

$$FeSO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + Fe(OH)_2(s)$$

9. First, dissolve the four white solids into distilled water to form four colourless solutions.

Treat each sample solution with dilute sodium hydroxide solution and then heat them. Only ammonium chloride and ammonium nitrate solution produce a gas which can turn wet red litmus paper blue.

$$NH_4^+(aq) + OH^-(aq) \rightarrow NH_3(g) + H_2O(l)$$

Treat the ammonium chloride solution and ammonium nitrate solution with silver nitrate solution and dilute nitric acid. Only ammonium chloride solution gives white precipitate (silver chloride).

$$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$$

Then treat sodium hypochlorite solution and sodium sulphate solution with barium nitrate solution. Only sodium sulphate solution gives white precipitate.

$$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$$

Note:
- The answer given above is only one of the possible answers in this question. In fact, there are many possible answers. You can get full content marks in this question if you can describe tests, including proper observations, to distinguish the four solids from one another.
- For identifying chloride ions, silver nitrate solution is often used because silver ions will react with chloride ions to form insoluble silver chloride, which appears as white precipitate.

  $$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$$

  However, both oxide and carbonate of silver are also insoluble in water. To eliminate these cases, dilute nitric acid is often added. Only silver carbonate and silver oxide will dissolve in dilute nitric acid.

  $$Ag_2O(s) + 2HNO_3(aq) \rightarrow 2AgNO_3(aq) + H_2O(l)$$

  $$Ag_2CO_3(s) + 2HNO_3(aq) \rightarrow 2AgNO_3(aq) + CO_2(g) + H_2O(l)$$

- For identifying sulphate ions, barium nitrate solution is often used because barium ions will react with sulphate ions to form insoluble barium sulphate, which appears as white precipitate.

  $$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$$

  However, both sulphite and carbonate of barium are also insoluble in water. To eliminate these cases, dilute hydrochloric acid is added. Only barium sulphite and barium carbonate will dissolve in dilute hydrochloric acid.

$$BaSO_3(s) + 2HCl(aq) \rightarrow BaCl_2(aq) + SO_2(g) + H_2O(l)$$

$$BaCO_3(s) + 2HCl(aq) \rightarrow BaCl_2(aq) + CO_2(g) + H_2O(l)$$

- For identifying hypochlorite ions, dilute acid is often used because hypochlorite ions will react with hydrogen ions to form chlorine gas, which is greenish yellow in colour.

$$Cl^-(aq) + OCl^-(aq) + 2H^+(aq) \rightarrow Cl_2(g) + H_2O(l)$$

PART II

10. (a) **Method 1**

	P(g)	Q(g)	R(g)
No. of moles at 0 min	0.50	0.50	0.00
No. of moles at 40 min	0.25	0.00	0.25

Hence, 1 mole of P(g) reacts with 2 moles of Q(g) to give 1 mole of R(g).

$\therefore$ $P(g) + 2Q(g) \rightarrow R(g)$

Method 2

The concentrations of P and Q decrease as they are reactants, while that of R increases as it is a product. The slope of concentration of P and that of R share the same magnitude at anytime, indicating that 1 mole of P is consumed to give 1 mole of R. Similarly, the slope of concentration of P is half of that of Q at anytime, so 1 mole of P reacts with 2 moles of Q.

(b) The rate of reaction depends on the concentration of the reactants in the system. With the reaction taking place in a closed container of increased volume, the concentration of reactants in the system decreases. There is a lower chance of the collision of reactants and the number of effective collisions decreases. The rate of reaction is slower and hence, it would take a longer time for the reaction to complete.

(c) The reaction would take place only when the reactants have enough energy to break the bonds in reactants to initiate the reaction, and they collide with correct orientation. Hence, the collisions between molecules of P(g) and Q(g) would not necessarily lead to a reaction.

11. (a) K_c increases when the temperature of the system increases. This shows that the equilibrium shifts to the right when the temperature of the system increases. As a higher temperature favours endothermic reactions, the forward reaction is endothermic.

(b)

	$H_2(g)$	+ $CO_2(g)$	$\rightleftharpoons$ $CO(g)$	+ $H_2O(g)$
Initial concentration ($mol\,dm^{-3}$)	0.5	0.5	0	0
Eqm. concentration ($mol\,dm^{-3}$)	$0.5 - x$	$0.5 - x$	x	x

$$\therefore \quad K_c = \frac{[CO(g)][H_2O(g)]}{[H_2(g)][CO_2(g)]}$$

$$1.23 \times 10^{-1} = \frac{(x)(x)}{(0.5-x)(0.5-x)}$$

$$0.123(x^2 - x + 0.25) = x^2$$

$$0.877x^2 + 0.123x - 0.03075 = 0$$

$$x = 0.130 \text{ or } -0.270 \text{ (rejected)}$$

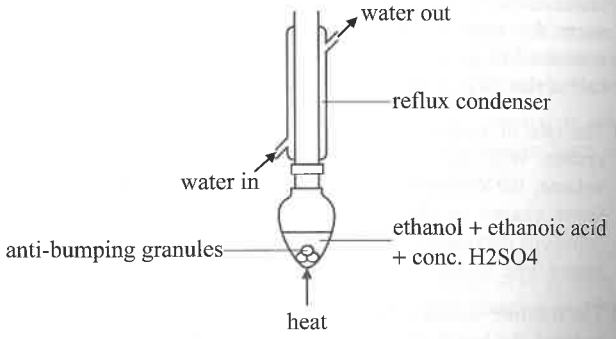
$\therefore$ The concentration of CO(g) in the equilibrium mixture is 0.130 mol dm^{-3}.

(c) An increase in temperature would lead to an increase in the rate of the backward reaction since the kinetic energy of reactants in the backward reaction increases.

12. (a) Concentrated sulphuric acid.

Note: Concentrated sulphuric acid serves as a dehydrating agent to remove water from the system, which shifts the equilibrium position of the system to the right and thus increases the rate of esterification.

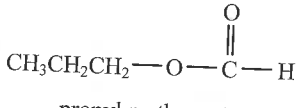
(b)



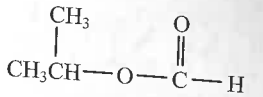
(c) Ethyl ethanoate's ability to dissolve iodine is due to the comparable intermolecular attraction between ethyl ethanoate and iodine molecules. Since both iodine and ethyl ethanoate molecules are held together by weak van der Waals' force, iodine can dissolve in ethyl ethanoate by 'like-dissolve-like' principle.

For sodium iodide, the electrostatic attraction between Na^+ and I^- resulting from the ionic bond is much stronger than the weak van der Waals' force between ethyl ethanoate molecules and thus, sodium iodide cannot be solvated by ethyl ethanoate.

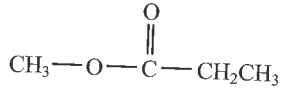
(d) (Any **ONE** of the following)



propyl methanoate

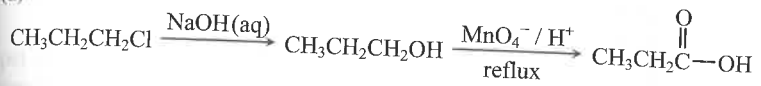


methylethyl methanoate

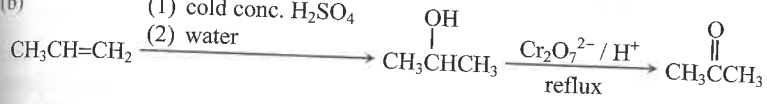


methyl propanoate

13. (a)



(b)



14. When Na_2O dissolves in water, the oxide ions react with water to give hydroxide ions, making the solution alkaline.

$$Na_2O(aq) + H_2O(l) \rightarrow 2Na^+(aq) + 2OH^-(aq)$$

When sulphur dioxide is dissolved in water, the sulphur which acquires a partial positive charge would be attacked by the lone pair electrons of the oxygen atom in water molecule, giving sulphurous acid.

$$SO_2(aq) + H_2O(l) \rightleftharpoons 2H^+(aq) + SO_3^{2-}(aq)$$

The acid formed may be ionized to give H^+ ions, so the resulting solution is acidic.

To summarize, Na_2O undergoes hydrolysis to give an alkaline solution while SO_2 reacts to give an acidic solution.

END

DSE Sample Paper CHEM P2

(1) Industrial Chemistry

(a) (i) $ClO_3^-(aq) + 5Cl^-(aq) + 6H^+(aq) \rightarrow 3Cl_2(g) + 3H_2O(l)$

(ii) **Method 1**

Initial rate of reaction increased by four times when concentration of $H^+(aq)$ is doubled while the concentration of $ClO_3^-(aq)$ and $Cl^-(aq)$ remain constant (as shown in the first and second row of the results). Hence, the order of reaction with respect to $H^+(aq)$ is 2.

Initial rate of reaction doubled when concentration of $ClO_3^-(aq)$ is doubled while the concentration of $Cl^-(aq)$ and $H^+(aq)$ remain constant (as shown in the second and third row of the results). Hence, the order of reaction with respect to $ClO_3^-(aq)$ is 1.

Initial rate of reaction doubled when concentration of $Cl^-(aq)$ is doubled while the concentration of $ClO_3^-(aq)$ and $H^+(aq)$ remain constant (as shown in the first and fourth row of the results). Hence, the order of reaction with respect to $Cl^-(aq)$ is 1.

Hence, we have $\text{rate} = k[ClO_3^-(aq)][Cl^-(aq)][H^+(aq)]^2$

where k is the rate constant of the reaction.

Method 2

Let the rate equation be $\text{rate} = k[ClO_3^-(aq)]^x[Cl^-(aq)]^y[H^+(aq)]^z$

where k is the rate constant of the reaction.

By substituting the given information into the rate equation, we have

$1.0 \times 10^{-5} = k(0.08)^x(0.15)^y(0.20)^z$ .....(1)

$4.0 \times 10^{-5} = k(0.08)^x(0.15)^y(0.40)^z$ .....(2)

$8.0 \times 10^{-5} = k(0.16)^x(0.15)^y(0.40)^z$ .....(3)

$2.0 \times 10^{-5} = k(0.08)^x(0.30)^y(0.20)^z$ .....(4)

(3) ÷ (2): $2 = 2^x \Rightarrow x = 1$

(4) ÷ (1): $2 = 2^y \Rightarrow y = 1$

(2) ÷ (1): $4 = 2^z \Rightarrow z = 2$

∴ The rate equation is $\text{rate} = k[ClO_3^-(aq)][Cl^-(aq)][H^+(aq)]^2$

where k is the rate constant of the reaction.

(iii) Using the information given from row 1,

$1.0 \times 10^{-5} = k(0.08)(0.15)(0.20)^2$

$k = 0.0208\ \text{mol}^{-3}\text{dm}^9\text{s}^{-1}$

(iv) By Arrhenius equation $k = Ae^{-\frac{E_a}{RT}}$

$$\ln\frac{k_2}{k_1} = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\ln 2 = \frac{E_a}{8.31}\left(\frac{1}{298} - \frac{1}{308}\right)$$

$E_a = +52.9\ \text{kJmol}^{-1}$

(b) In ammonia plant, purified and dried nitrogen and hydrogen are mixed in a ratio of 1 to 3 and compressed at 200 atm to 250 atm. The mixture is heated at 450°C to 500°C in the heat exchanger. The hot gaseous mixture is then passed over finely divided and enriched iron powder in the catalytic converter. Ammonia is produced from the following reaction at a yield of 15%:

$$N_2(g) + 3H_2(g) \underset{\text{Fe catalyst}}{\overset{\text{723 K, 200 atm}}{\rightleftharpoons}} 2NH_3(g)$$

Ammonia synthesized from the Haber Process is liquefied and stored in the storage tank as liquid ammonia. The liquid ammonia synthesised is then transported to the nitric acid plant.

Nitric acid is then manufactured by the catalytic oxidation of ammonia. The raw materials involved are ammonia, air and water. And the following reactions are involved:

Firstly, $NH_3(g)$ is catalytically oxidised to $NO(g)$.

$$4NH_3(g) + 5O_2(g) \xrightarrow[\text{Pt catalyst}]{} 4NO(g) + 6H_2O(l)$$

$NO(g)$ then reacts with atmospheric oxygen to $NO_2(g)$.

$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

Eventually, $NO_2(g)$ formed is dissolved in water in the presence of air to form nitric acid.

$4NO_2(g) + O_2(g) + 2H_2O(l) \rightarrow 4HNO_3(aq)$

(c) (i) Atom economy

$$= \frac{\text{molar mass of atoms in the desired product}}{\text{sum of molar mass of all products}} \times 100\%$$

$$= \frac{12.0\times4+1.0\times8}{(12.0\times4+1.0\times8)+(12.0\times2+1.0\times6+16.0)+(23.0+79.9)} \times 100\%$$

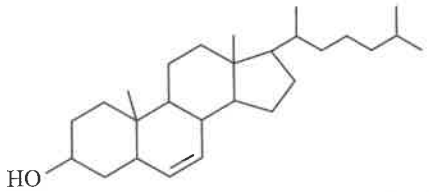
$$= 27.3\%$$

(ii) (Any **TWO** of the following)

- waste prevention at source
- designing for energy efficiency
- using renewable raw materials
- using catalysts
- using a solvent-free system or safer and benign solvent system
- using less hazardous chemical syntheses
- producing safer chemical products
- reducing derivatives
- designing degradable chemical products

(2) Materials Chemistry

(a) (i)



(ii) True liquid exhibits distinctive phase while liquid crystals exhibit phase intermediate between solid and liquid. Molecules in 'true liquid' have no intrinsic order while molecules in liquid crystal are arranged orderly having certain degree of translational freedom with certain orientation preference, as in the case of a solid crystal, in the liquid-crystalline phase. A liquid crystal can exhibit one or more ordered phases e.g. nematic, smectic and cholesteric phase at a temperature above the melting point of the solid.

(iii) The chiral nematic phase arises when the molecules that make up the nematic phase are chiral. In the cholesteric phase, the liquid crystal molecules are arranged in a similar manner as in nematic phase. The molecules are arranged orderly in each layer. Neighbouring layers have molecules pointing at different angles. The molecules line up in positions in such a way that they are slightly twisted from the molecules in the neighbouring layer. This results in the helical arrangement of the molecules.

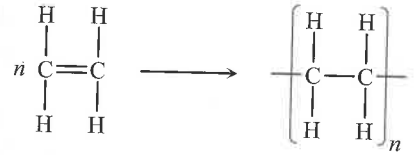
(iv) A LCD contains liquid crystal molecules in cholesteric phase sandwiched between a polariser and an analyser.

When ordinary light passes through the first polariser that allows only light in one plane to pass through, the plane of polarisation of the polarised light is rotated 90° by the liquid crystal molecules that are naturally twisted and are pointing to their original orientations. Thus, the polarised light is in the correct orientation to pass through the second polariser and hence the analyser .So the display area appears bright.

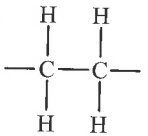
When a voltage is applied across the two glass electrodes, the liquid crystal molecules reorient parallel to the electric field. The plane of polarised light is not rotated by the molecules. In that case, the light cannot pass through the second polariser. The display appears dark.

(b) (i) Cross-links are formed between the polymer chains when a thermosetting plastic is moulded and a hard and rigid three-dimensional giant network is thus formed , whereas there are no cross-links between polymer chains in thermoplastics such that only weak intermolecular forces are holding polymer chains to one another and the polymeric chains are oriented randomly. Hence, the weak Van der Waals' force between polymer chains can easily be overcome by heating and the polymer chains in thermoplastics can slide over one another easily while the polymer chains in thermosetting plastics cannot slide over one another even when heated owing to the rigid giant network. As a result, thermoplastics would soften while thermosetting plastics would not softened upon heating.

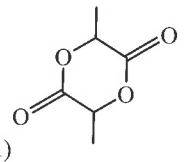
(ii) (I)



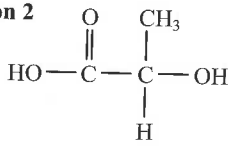
(II)



(iii)(I) **Solution 1**



Solution 2



(II) **(Solution 1)**
3,6-dimethyl-l,4-dioxan-2,5-dione

(Solution 2)
2-hydroxypropanoic acid

(III) **(Solution 1)**
ring-opening polymerisation

(Solution 2)
condensation polymerisation

(IV) PLA is synthesised directly from renewable natural resources as lactic acid which is the precursor that can be generated from fermentation of corn starch or sugar(which forms from the photosynthesis of corn using carbon dioxide and water)such that it can be broken down by the enzymatic action of micro-organisms in nature easily (biodegradable). Also, they can be easily hydrolysed by acid or alkali.

(3) Analytical Chemistry

(a) (i) Consider the following table:

	Carbon	Hydrogen	Oxygen
Mass ratio	0.621	0.103	0.276
Mole ratio	$\frac{0.621}{12.0} = 0.05175$	$\frac{0.103}{1.0} = 0.103$	$\frac{0.276}{16.0} = 0.01725$
Simplest ratio	$\frac{0.05175}{0.01725} = 3$	$\frac{0.103}{0.01725} \approx 6$	$\frac{0.01725}{0.01725} = 1$

$\therefore$ The empirical formula of **G** is C_3H_6O.

Let the molecular formula of **G** be $(C_3H_6O)_n$, where n is an integer.

$\therefore \quad n(12.0 \times 3 + 1.0 \times 6 + 16.0) = 58$

$n = 1$

$\therefore$ The molecular formula of **G** is C_3H_6O.

(ii) According to the infrared spectrum, there is a prominent dip at wavenumber 1700 cm^{-1}, showing that a carbonyl (C=O) group is present in **G**. Hence, **G** is a carbonyl compound.

According to the mass spectrum, intensity peak at m/z ratio = 58 corresponds to the molecular ion peak which is formed by $[CH_3CH_2CHO]^+$. There is also another prominent intensity peak at m/z ratio = 29 corresponding to the ions $[CHO]^+$ or $[C_2H_5]^+$. So a terminal carbonyl group is present in **G**. Thus, **G** cannot be propanone. Hence, **G** should be propanal.

$\therefore$ The structural formula of **G** is CH_3CH_2CHO.

(iii) Treat **G** with 2,4-dinitrophenylhydrazine to prepare the corresponding 2,4-dinitrophenylhydrazone. Recrystallize the 2,4-dinitrophenylhydrazone. The melting point of the solid derivative is determined using a melting point apparatus. Then, a sharp melting point which is the same as that quoted in the literature value would support the deduction of **G**.

(b) (i) $3CH_3CH_2OH + 2Cr_2O_7^{2-} + 16H^+ \rightarrow 3CH_3COOH + 4Cr^{3+} + 11H_2O$

(ii) (Any **TWO** of the following)
- low hygroscopy
- low efflorescence
- large relative molecular mass
- high purity
- high stability in air

Note: A primary standard in titrimetric analysis is a pure reagent from which its solution of accurately known molarity can be directly prepared by dissolving a known mass of the solid and used for titrimetric analysis.

(iii) Keep a small sample for 2 days. Repeat the titration process to see if the result is the same. If so, the reaction has completed.

(iv) $Fe^{2+}(aq)$ present in $(NH_4)_2Fe(SO_4)_2(aq)$ would reduce unreacted $Cr_2O_7^{2-}(aq)$ to $Cr^{3+}(aq)$.

$6Fe^{2+}(aq) + Cr_2O_7^{2-}(aq) + 14H^+(aq) \rightarrow 6Fe^{3+}(aq) + 2Cr^{3+}(aq) + 7H_2O(l)$

No. of moles of $(NH_4)_2Fe(SO_4)_2 = 0.118 \times (12.23 \times 10^{-3}) = 1.44314 \times 10^{-3}$

Mole ratio of $Fe^{2+} : Cr_2O_7^{2-} = 6 : 1$

No. of moles of excess $Cr_2O_7^{2-}$ present
$= 1.44314 \times 10^{-3} \div 6$
$= 2.40523 \times 10^{-4}$

No. of moles of $Cr_2O_7^{2-}$ reacted
$= (25.0 \times 10^{-3}) \times 0.156 - (2.40523 \times 10^{-4})$
$= 3.65948 \times 10^{-3}$

Mole ratio of $CH_3CH_2OH : Cr_2O_7^{2-} = 3 : 2$

No. of moles of ethanol reacted $= (3.65948 \times 10^{-3}) \times \frac{3}{2} = 5.48922 \times 10^{-3}$

Molarity of ethanol in the 25.0 cm^3 diluted spirit
$= \frac{5.48922 \times 10^{-3}}{25.0 \times 10^{-3}}$
$= 0.219569$ M

As the spirit is diluted by 25 times,

Molarity of ethanol in the spirit $= 0.219569 \times 25 = 5.49$ M

$\therefore$ Concentration of ethanol in the brand of spirit is 5.49 M.

END